

Name: Solution

MATH 241

QUIZ#4

SUMMER 2014

Problem 1. (5 Points)

Let $\mathcal{X} \equiv C^0[-1,1]$ be the vector space of real valued continuous functions defined on the interval $[-1,1]$, equipped with the dot (inner) product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x)dx \quad f, g \in \mathcal{X} \text{ and the induced norm } \|f\| = \sqrt{\langle f, f \rangle}.$$

Define the vector space $S \equiv \text{Span}\{f, g\}$, where $f(x) = \cos(\pi x)$ and $g(x) = \sin(\pi x)$.

- Is $\cos(\pi x + \frac{\pi}{3})$ in S ?
- Verify that f and g form an orthonormal basis for S .
- Find the projection of h onto S where $h(x) = \cos(x)$.

① a) $\cos(\pi x + \frac{\pi}{3}) = \cos(\frac{\pi}{3})\cos(\pi x) - \sin(\frac{\pi}{3})\sin(\pi x) \rightarrow$ full credit for a plus between terms!

$$= \frac{1}{2}\cos(\pi x) - \frac{\sqrt{3}}{2}\sin(\pi x) = \frac{1}{2}\underline{f} + \left(-\frac{\sqrt{3}}{2}\right)\underline{g}$$

$$\Rightarrow \cos(\pi x + \frac{\pi}{3}) \in S$$

① b) $\{\underline{f}, \underline{g}\}$ is an orthonormal basis then $\boxed{\|\underline{f}\|=1}$ $\boxed{\|\underline{g}\|=1}$ $\boxed{\langle \underline{f}, \underline{g} \rangle = 0}$

full credit if the integration is correctly written. No need to evaluate them.

$$\int \|\underline{f}\|^2 = \int_{-1}^1 \cos^2(\pi x) dx = \int_{-1}^1 \frac{\cos(2\pi x) + 1}{2} dx = \underbrace{\int_{-1}^1 \frac{\cos(2\pi x)}{2} dx}_{0 \text{ (integration over one period!)}} + \int_{-1}^1 \frac{1}{2} dx = 1$$

$$\|\underline{g}\|^2 = \int_{-1}^1 \sin^2(\pi x) dx = \int_{-1}^1 \frac{1 - \cos(2\pi x)}{2} dx = 1$$

$$\langle \underline{f}, \underline{g} \rangle = \int_{-1}^1 \sin(\pi x) \cos(\pi x) dx = \int_{-1}^1 \frac{\sin(2\pi x)}{2} dx = \underbrace{-\frac{\cos(2\pi x)}{4\pi}}_{0 \text{ (integration over two periods!)}} \Big|_{-1}^1 = -\frac{1}{4\pi} (\cos(2\pi) - \cos(-2\pi)) = 0$$

① writing this relation.

$$c) \underline{h}_p = \langle \underline{f}, \underline{h} \rangle \underline{f} + \langle \underline{g}, \underline{h} \rangle \underline{g} = \frac{2\sin(1)}{\pi^2-1} \cos(\pi x) + 0 \cdot \sin(\pi x) = \frac{2\sin(1)}{\pi^2-1} \cos(\pi x)$$

writing integrals

$$\langle \underline{g}, \underline{h} \rangle = \int_{-1}^1 \underbrace{\sin(\pi x)}_{\text{odd}} \underbrace{\cos(x)}_{\text{even}} dx = \int_{-1}^1 \{\text{odd function}\} dx = 0 \quad \text{since the integration of odd functions is zero over an interval } [-a, a]$$

① for evaluating integration:

$$\langle \underline{f}, \underline{h} \rangle = \int_{-1}^1 \cos(\pi x) \cos(x) dx = \frac{1}{2} \int_{-1}^1 [\cos((\pi+1)x) + \cos((\pi-1)x)] dx = \frac{1}{2} \left[\frac{\sin((\pi+1)x)}{\pi+1} + \frac{\sin((\pi-1)x)}{\pi-1} \right] \Big|_{-1}^1$$

$$= \frac{1}{2} \left[\frac{\sin(\pi+1) - \sin(-(\pi+1))}{\pi+1} + \frac{\sin(\pi-1) - \sin(-(\pi-1))}{\pi-1} \right] = \frac{1}{2} \left[\frac{2\sin(1)}{\pi+1} + \frac{2\sin(1)}{\pi-1} \right] = \frac{2\sin(1)}{\pi^2-1}$$

Problem 2. (5 Points)

Let S be a vector space defined as a subspace of $\mathbb{R}^3$ consisting of all the points lying on the plane defined by

$$x_1 - x_2 + 2x_3 = 0 \text{ for } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in S.$$

(a) Determine a basis for S .

(b) Find a vector $\mathbf{w} \in \mathbb{R}^3$ which is orthogonal to S .

③ a)
$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \text{pivot} & \text{free variables} & \end{array} \quad \begin{bmatrix} 1 & -1 & 2 & : & 0 \end{bmatrix} \rightarrow \begin{array}{l} x_2 = \alpha \\ x_3 = \beta \end{array}$$

$$\mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Since the plane is in $\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$ which forms a basis, since the vectors in the generating set is LI.

② b) Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ which is ^{formed from} the coefficients of x_1, x_2, x_3 in the

equation $x_1 - x_2 + 2x_3 = 0$, this equation can be written in the form $\mathbf{u}^T \mathbf{x} = 0 = \mathbf{u} \cdot \mathbf{x}$ since $\mathbf{x}$ and $\mathbf{u}$ are orthogonal.

are orthogonal, then $\mathbf{w} = \mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

verification
$$\mathbf{x} \cdot \mathbf{u} = \left(\alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right) \cdot \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \boxed{\alpha \cdot 0 + \beta \cdot 0 = 0}$$